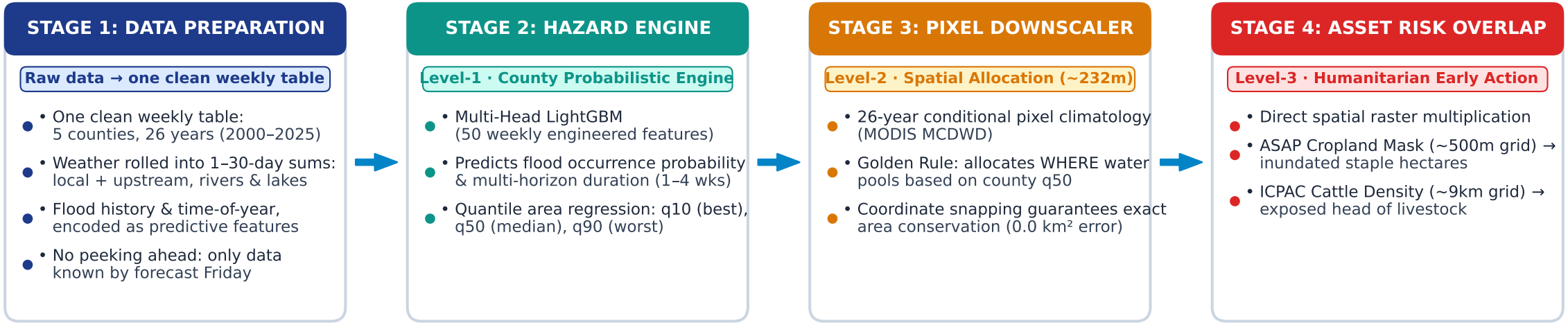
STAGE 1: DATA PREPARATION
Raw data → one clean weekly table
• One clean weekly table: 5 counties, 26 years (2000–2025)
• Weather rolled into 1–30-day sums: local + upstream, rivers & lakes
• Flood history & time-of-year, encoded as predictive features
• No peeking ahead: only data known by forecast Friday
STAGE 2: HAZARD ENGINE
Level-1 · County Probabilistic Engine
• Multi-Head LightGBM (50 weekly engineered features)
• Predicts flood occurrence probability & multi-horizon duration (1–4 wks)
• Quantile area regression: q10 (best), q50 (median), q90 (worst)
STAGE 3: PIXEL DOWNSCALER
Level-2 · Spatial Allocation (~232m)
• 26-year conditional pixel climatology (MODIS MCDWD)
• Golden Rule: allocates WHERE water pools based on county q50
• Coordinate snapping guarantees exact area conservation (0.0 km² error)
STAGE 4: ASSET RISK OVERLAP
Level-3 · Humanitarian Early Action
• Direct spatial raster multiplication
• ASAP Cropland Mask (~500m grid) → inundated staple hectares
• ICPAC Cattle Density (~9km grid) → exposed head of livestock